

Ex.no:5b	DATA VISUALIZATION USING MATPLOTLIB – SALES DATA

AIM

To visualize and analyze the Vehicle Analysis dataset using Line Chart, Bar Chart, and Pie Chart with Matplotlib to understand attack trends, regional comparisons, and attack distribution.

ALGORITHM

Step 1: Import Pandas and Matplotlib libraries.

Step 2: Load the Vehicle Analysis dataset into a DataFrame.

Step 3: Group the vehicle market according to year and calculate the number of vehicles for each year.

Step 4: Plot the yearly sales count using a Line Chart to observe the trend of vehicle market over time.

Step 5: Group the dataset according to region and calculate the number of fuel in each region.

Step 6: Plot the regional sales counts using a Bar Chart to compare vehicle market sales across different regions.

Step 7: Select the major regions and calculate their sales proportions.

Step 8: Plot the regional distribution using a Pie Chart.

Step 9: Add suitable titles, axis labels, legends, and display the visualizations.

CODE

```
import pandas as pd
import matplotlib.pyplot as plt
import kagglehub
import os

# Load Vehicle Dataset from Kaggle
path = kagglehub.dataset_download(
    "nehalbirla/vehicle-dataset-from-cardekho"
)
```

```
print("Dataset Path:")
print(path)
# Find CSV file automatically
csv_files = [f for f in os.listdir(path) if f.endswith(".csv")]
csv_file = csv_files[0]
# Load Dataset
df = pd.read_csv(
    os.path.join(path, csv_file),
    encoding="latin1",
    low_memory=False
)
# Clean column names
df.columns = df.columns.str.strip().str.lower().str.replace(" ", "_")
print("\nDataset Loaded Successfully")
print("\nColumn Names:")
print(df.columns.tolist())
# -----
# 1. Line Chart – Vehicles by Year
# -----
yearly_vehicles = df.groupby("year").size()
plt.figure(figsize=(10, 5))
plt.plot(
    yearly_vehicles.index,
    yearly_vehicles.values
)
plt.title("Vehicles by Year")
plt.xlabel("Year")
plt.ylabel("Number of Vehicles")
plt.grid(True)

plt.show()
```

```

# -----
# 2. Bar Chart – Vehicles by Fuel Type
# -----

fuel_vehicles = df["fuel_type"].value_counts()
plt.figure(figsize=(10, 6))
plt.bar(
    fuel_vehicles.index,
    fuel_vehicles.values
)
plt.title("Vehicles by Fuel Type")
plt.xlabel("Fuel Type")
plt.ylabel("Number of Vehicles")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.show()

# -----
# 3. Pie Chart – Fuel Type Distribution
# -----

top_fuels = fuel_vehicles.head(6)
plt.figure(figsize=(8, 8))
plt.pie(
    top_fuels.values,
    labels=top_fuels.index,
    autopct="%1.1f%"
)
plt.title("Fuel Type Distribution of Vehicles")
plt.show()

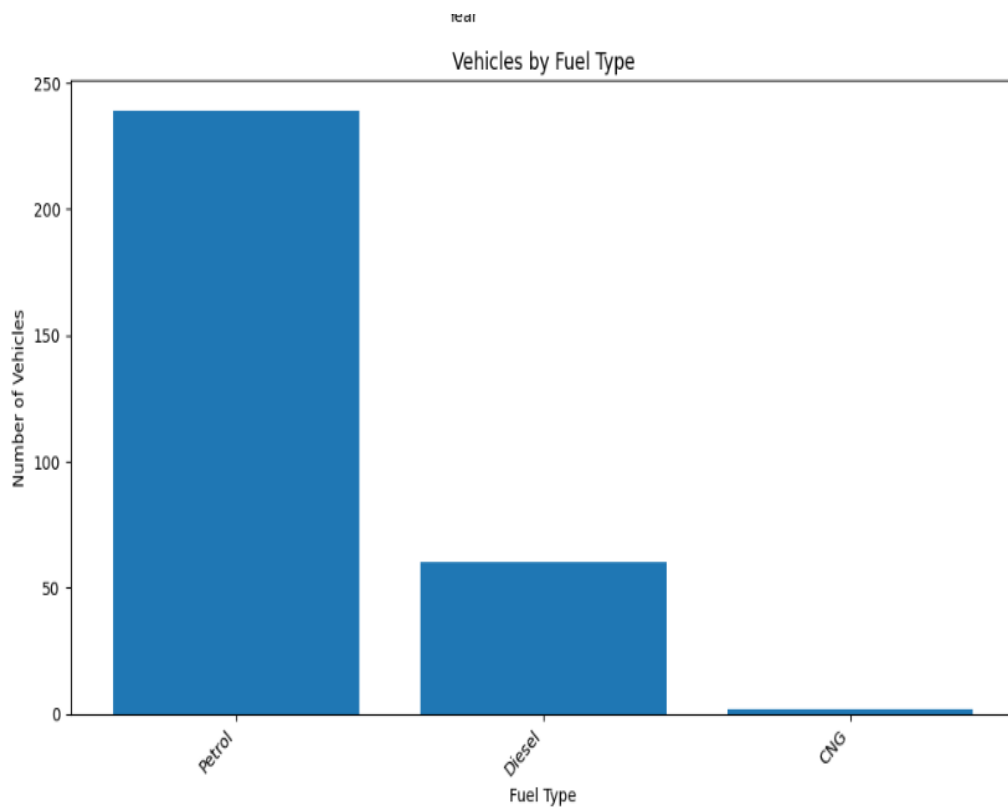
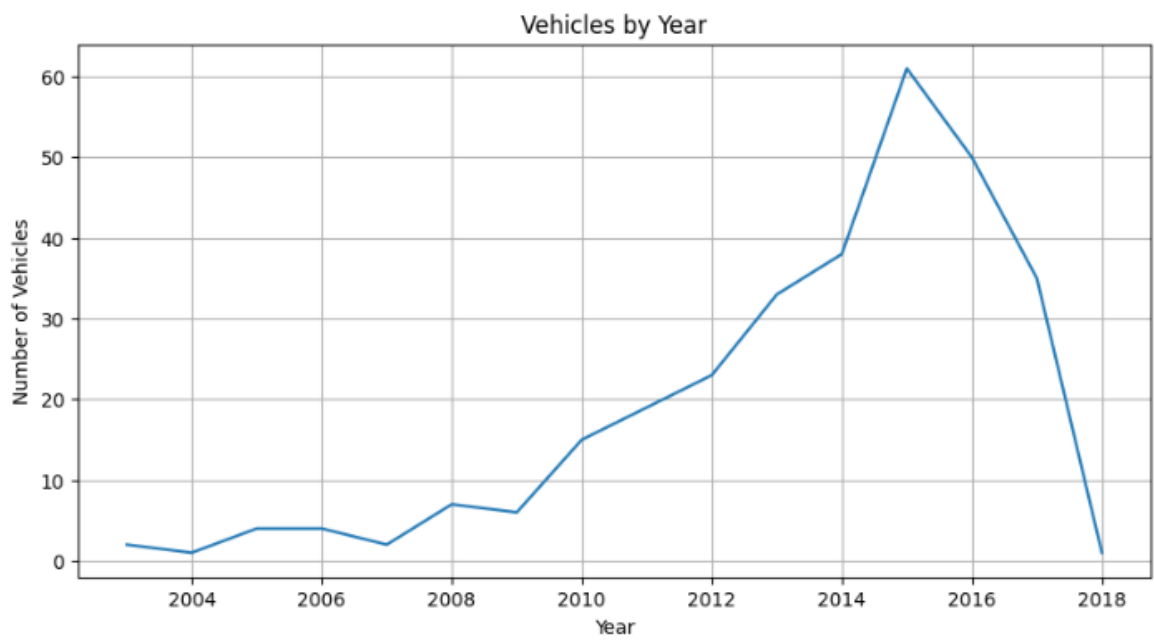
```

OUTPUT

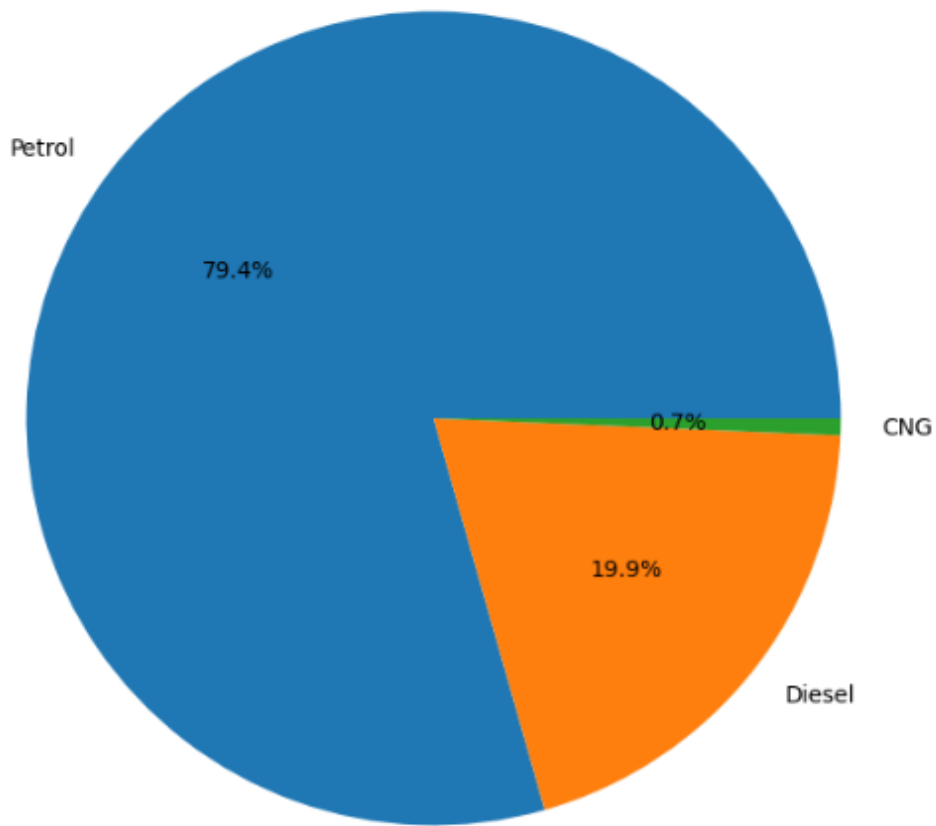
Dataset Loaded Successfully

Column Names:

```
['car_name', 'year', 'selling_price', 'present_price', 'kms_driven', 'fuel_type', 'seller_type', 'transmission', 'owner']
```



Fuel Type Distribution of Vehicles



RESULT